

NATHANIEL McNEILL

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EDUCATION

University of California-San Diego	Bachelor of Science, Data Science and Cognitive Science (Specialization in Machine Learning and Neural Computation) GPA: 3.28/4.0	June 2028
Highland Park High School	Talented and Gifted Program Multidisciplinary Studies and Distinguished LLevel of Achievement Summa Cum Laude GPA: 4.389/4.0	May 2024

PROJECTS

Mini Data Science Framework - UCSD DSC 30 Quarter-Long Project GitHub Repo	Spring 2025
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- Built a DataFrame class in Java with linked lists and binary search for efficient data handling.
- Engineered a generic hash table with collision handling and dynamic resizing.
- Created Series (1-Dimensional) and DataFrame (2-Dimensional) structures supporting indexing and column operations.
- Applied object-oriented principles to ensure scalability and maintainability.

TV Sitcom Data Analysis Project - UCSD DSC 10 Final Project GitHub Repo	Fall 2024
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- Analyzed Friends episode, script, and sentiment datasets (7k+ lines, 10 seasons) using Python in Jupyter Notebook
- Cleaned and transformed nested string fields to evaluate prolific writers/directors across 236 episodes.
- Applied bootstrapping and hypothesis testing to compare directors' IMDb ratings and episode viewership with 99% confidence intervals.
- Conducted gender balance analysis on 6,500+ main-character lines, testing for significance in dialogue distribution.
- Implemented a 2-gram Markov text generator to create new, realistic episode titles based on existing episode titles.

WORK EXPERIENCE

UP Mustangs Basketball Academy - Assistant Coach ; Dallas, TX	Winter 2022 - Summer 2024
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- Developed and implemented strategic training drills to address team weaknesses
- Established individual player goals and mentored athletes to improve performance
- Communicated with parents to support their child's growth and prepare for competitive play

LEADERSHIP EXPERIENCES AND ACTIVITIES

Young Men's Service League - Park Cities Chapter ; Dallas, TX	Fall 2018 - Spring 2024
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- Served as Life Skills, Leadership, and Slating Chair, leading workshops and mentoring members.
- Organized training sessions and directed transparent leadership selection processes for 100+ members.

HONORS AND AWARDS

• Provost Honors (3 quarters) • National Merit Finalist	Fall 2024 - Spring 2025 Fall 2023
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ADDITIONAL INFORMATION

Soft Skills: Attention to Detail, Critical Thinking, Communication, Adaptability

Software Skills: Jupyter Notebook, Excel, MATLAB, Sheets, LaTeX

Computer Languages: Python and Java

Mathematical Skills: Linear Algebra, Multivariable Calculus, Statistical Methods

Other Academic Skills: Physics, Experimental Design

Certifications: UT Computer Science Academy for iOS App Development (2023)

Interests: Learning, Travelling, Human & Artificial Cognition, Data Analytics

Work Eligibility: Eligible to work in the U.S. with no restriction